Emmanuel Alabi

Software Engineer & AI/Robotics Researcher

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Research Interests

Embodied AI; robot learning; computer vision; autonomous systems; multimodal intelligence. Specifically, how an agent's representation of a physical environment should be shaped by having to move through it, and when learned components genuinely outperform well-specified classical methods.

Broader interests: machine learning, robotics, state estimation, planning, reinforcement learning.

Education

Babcock University — BSc (Hons) Software Engineering 2019 — 2023

Second Class Upper Division, CGPA 4.00/5.00. Ilishan-Remo, Ogun State, Nigeria. School of Computing and Engineering Sciences.

Relevant coursework: Algorithms and Data Structures; Discrete Mathematics; Introductory Statistics; Introduction to Operations Research; Artificial Intelligence and Applications; Introduction to Big Data Engineering; Computer Organization and Assembly Language; Operating System I; Database System Design, Implementation and Management.

Research

Risk-Calibrated Semantic Navigation Under Distribution Shift 2026

Preregistered ROS 2 / Gazebo / Nav2 benchmark. doi.org/10.5281/zenodo.22181941

- Calibration improved and clean-route efficiency held, but the preregistered collision-reduction and severity hypotheses did not: better component calibration did not establish safer navigation.
- Four systems differing only in the treatment component; protocol, manifest, thresholds and analysis code hash-locked before execution. 6,840 episodes; independent clean-clone reproduction byte-identical.
- emmanuelalabi.com/research/risk-calibrated-semantic-navigation

Driver Drowsiness Detection — Reproduction & Evaluation Study 2026

Independent computer vision study. github.com/alabiemmanuel177/driver-drowsiness-reproduction

- An oracle-classifier decomposition showed the historical temporal threshold, not residual classification error, dominated false alerts once classifier accuracy was sufficient.
- Subject-disjoint reproduction on the MRL Eye Dataset; CNN trained from scratch against a HOG+SVM baseline, with deterministic splits, automated leakage checks and three-seed ablations.
- emmanuelalabi.com/projects/driver-drowsiness-reproduction

Research Training

Independent Research Programme — Embodied & Multimodal Autonomy 2026 — present

- Self-directed programme in the mathematical, machine-learning and robotics foundations for independent research: linear algebra, probability, optimisation, state estimation and rigid-body geometry, each implemented from first principles.
- Reproducing published baselines in perception and navigation before extending them.
- emmanuelalabi.com/research/research-programme-2026

Selected Academic Projects

BUCODEL — Learning Management System for Open & Distance e-Learning 2022 — 2023

Undergraduate Final-Year Project (SENG490, 72/100), Babcock University. Supervisor: Prof. Stephen Maitanmi.

- Co-designed and implemented a web-based learning management system for Babcock University's Centre for Open and Distance e-Learning, supporting student, lecturer and administrator workflows.
- Built course and resource management, assignments, authentication, user administration, collaborative course activity and role-specific interfaces using React, Node.js/Express and MongoDB.
- Conducted requirements analysis, system design, iterative implementation under a spiral methodology, and interface/usability, unit and integration testing.
- emmanuelalabi.com/projects/bucodel-learning-management-system

Selected Technical Experience

Founder & CEO — Obelo Oct 2025 — present

- Production AI platform: multi-agent orchestration over a structured brand representation, retrieval-augmented context, model routing across model families, asynchronous orchestration, and an automated brand-consistency grader checking every output before release.
- emmanuelalabi.com/projects/obelo-multimodal-ai-infrastructure

Software Engineer — Punch Group Jan 2025 — Jul 2025

- Shipped three production applications: GraphQL optimisation and memory-leak resolution on a Next.js/Express product, a lead-generation tool with HubSpot and Salesforce integrations, and a real-time support service with in-app video.

Software Engineer — SPay Business Nov 2023 — Dec 2024

- Production mobile-financial systems: authentication improvements, application performance, interface redesign, CI/CD automation and over-the-air mobile deployments.

Software Engineer — Union Bank of Nigeria Jul 2023 — Nov 2023

- Enterprise applications in React and Java; designed and enforced website security protocols; delivered merchant notification systems over REST APIs.

Founder — Acadu Feb 2022 — Sep 2025

- Educational platform across mobile and web. Led product, architecture and engineering: React Native, Next.js and NestJS over PostgreSQL, deployed on Fly.io and GCP with Redis and RabbitMQ for caching, message queuing and real-time event processing. Paused 2025.

Head of Technology and Digital Strategy — Royal Gate Group May 2018 — present

Part-time, alongside study and subsequent engineering roles.

- Eight years leading technology across a multi-sector group of ten-plus subsidiaries: group-wide digital infrastructure, IT and network systems, and ongoing digital transformation programmes including e-commerce platforms and internal systems.

Technical Skills

Research computing Python, PyTorch, NumPy, ROS 2, Gazebo, Nav2, statistical evaluation, uncertainty calibration, experiment design

AI systems Computer vision, multimodal systems, retrieval-augmented generation, LLM and model integration, evaluation systems

Software & infrastructure TypeScript, SQL, Docker, CI/CD, PostgreSQL, MongoDB, Node.js, React, Next.js

References

Prof. Stephen Maitanmi

Professor and Head of Department, Software Engineering, Babcock University. Supervised the assessed final-year research project (SENG490, 2023).

Mr. Akinwale Ojo

Founder and Chief Executive Officer, SPay Business Solutions Ltd. Software engineering role, November 2023 to December 2024.

Contact details available on request.